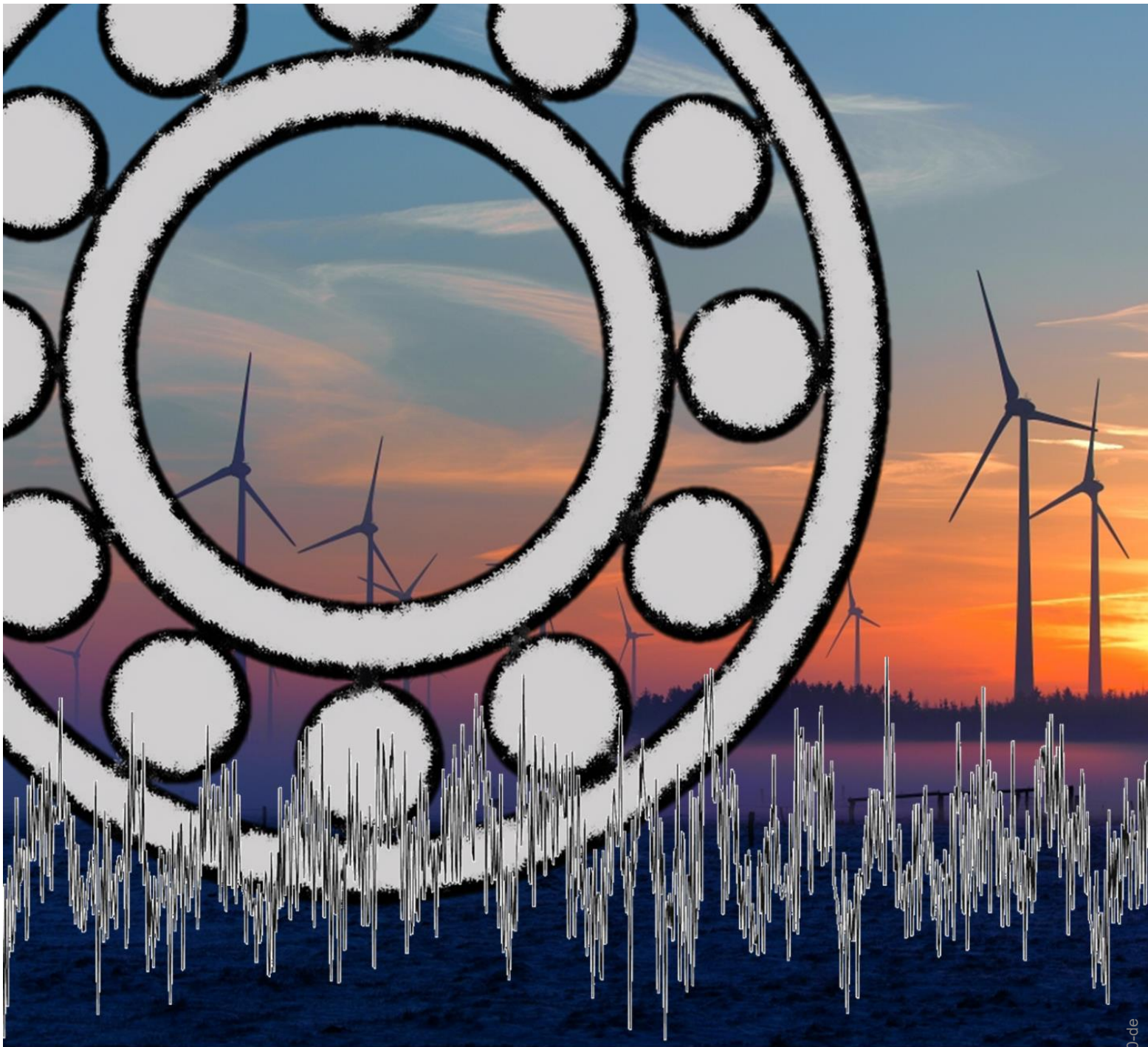




Minimum requirements for portable rotor bearing measurements on ENERCON wind energy converters

Publisher ENERCON Global GmbH ▪ Dreekamp 5 ▪ 26605 Aurich ▪ Germany
Phone: +49 4941 9270 ▪ Fax: +49 4941 927109
Email: info@enercon.de ▪ Internet: <http://www.enercon.de>
Managing Directors: Uwe Eberhardt, Heiko Juritz
Local court: Aurich ▪ Company registration number: HRB 202549
VAT ID no.: DE285537483

Copyright notice The entire content of this document is protected by copyright and – with regard to other intellectual property rights – national and international laws and treaties. ENERCON Global GmbH holds the rights in the content of this document unless another rights holder is expressly identified or obviously recognisable.

ENERCON Global GmbH grants the user the right to make copies and duplicates of this document for informational purposes for its own intra-company use; making this document available does not grant the user any further right of use. Any other duplication, modification, dissemination, publication, circulation, surrender to third parties and/or utilisation of the contents of this document – also in part – shall require the express prior written consent of ENERCON Global GmbH unless any of the above is permitted by mandatory legislation.

The user is prohibited from registering any industrial property rights in the know-how reproduced in this document, or for parts thereof.

If and to the extent that ENERCON Global GmbH does not hold the rights in the content of this document, the user shall adhere to the relevant rights holder's terms of use.

Registered trademarks Any trademarks mentioned in this document are intellectual property of the respective registered trademark holders; the stipulations of the applicable trademark law are valid without restriction.

Reservation of right of modification ENERCON Global GmbH reserves the right to change, improve and expand this document and the subject matter described herein at any time without prior notice, unless contractual agreements or legal requirements provide otherwise.

Area of application

This document applies to: International

Document details

Document ID	D03220088/0.0-en
Confidentiality	RESTRICTED

Date	Language	DCC	Plant/Department
2026-01-02	en	DC	Service Deutschland GmbH/Mechanical Engineering

Revision table

Revision	Chapter	Change
0.0	-	Document created

Table of contents

1	General information	5
2	Safety instructions	6
3	Qualifications of personnel	6
3.1	Measurement personnel on wind energy converters	6
3.2	Analysis personnel	6
4	The measuring equipment	7
4.1	Sensors	7
4.2	Measuring equipment (portable system)	7
5	Measurement	7
5.1	Operating conditions during measurement	7
5.2	Wind energy converter and bearing-specific data.....	8
5.3	Installation and coupling of measurement system.....	9
5.4	Dismantlement of the measurement system	11
6	Evaluation.....	12
6.1	Analysis of recorded vibration signals	12
7	Data required for the diagnostic report	13
8	Checklist for the minimum requirements	14

1 General information

This document describes the procedure for performing mobile vibration measurements on the rotor bearings of ENERCON wind energy converters.

The aim of the document is to ensure representative and comparable results of vibration measurements. Other methods of measurement or types of measuring equipment that do not satisfy the minimum requirements stated here might not be processed by ENERCON.

Attention must be given to the contents of analysis reports so that the reports can be evaluated.

Reporting of an error or an irregularity on the basis of a single measurement will not be accepted by ENERCON as a rule and will be rejected. A minimum of 3 measurements are required, which must be performed at an interval appropriate to the irregularity (e.g. 3 months).

Measurements should be carried out at a minimum of $\frac{3}{4}$ of the nominal speed.

Photos of the installation should also be taken and attached to the analysis report.

You are welcome to agree appropriate measurements with ENERCON in this respect.

Be sure to read and understand the general safety instructions before commencing work. The notes in the respective operating manuals and the requirements of national/international law and GWO standards apply.

The images in these instructions are for illustrative purposes only and can differ from the actual components present.

DIN ISO 10816-21 'Mechanical vibration - Horizontal axis wind turbines with gearbox' is **not** applied to ENERCON wind energy converters!

ENERCON reserves the right to adapt the content of this document on a regular basis in respect of new discoveries and the state of the art.

ENERCON will also continue to examine individual analysis reports critically and, if applicable, to perform counter-analyses to such analysis reports.

2 Safety instructions

Improper conduct in or around the WEC

Operating a wind energy converter involves certain risks.

- ⇒ Read the operating instructions of the wind energy converter carefully.
- ⇒ Observe all notes in the operating instructions of the wind energy converter, especially the safety instructions.
- ⇒ Refrain from any activities in and on the wind energy converter that could have unforeseeable effects or that are prohibited by the operating instructions.
- ⇒ Observe the additional safety provisions applicable at the deployment site.

Danger from idling rotor

Performing work while the rotor is not locked may cause serious injury or death.

- ⇒ Do not enter the rotor area unless the rotor lock is set.
- ⇒ Do not work on or near the rotor unless the rotor lock is set.
- ⇒ Before setting the rotor lock, check whether setting the rotor lock is permissible.
- ⇒ Before releasing the rotor lock, ensure that there are no persons near the rotor.
- ⇒ Keep a safe distance from rotating parts.

As a rule: The general safety instructions for the WEC apply. Observe these under all circumstances.

3 Qualifications of personnel

Performing the diagnostic of the rolling-element bearings by vibration analysis is a complex procedure that requires specialist knowledge. The personnel involved in measurement and evaluation of vibration analyses for the rotor bearings of ENERCON wind energy converters must possess the following qualifications:

3.1 Measurement personnel on wind energy converters

Person 1	Person 2
Vocational training in a technical field	Vocational training in a technical field
ENERCON Basic Training	ENERCON Basic Training
Ability to work at height (personal safety logbook)	Ability to work at height (personal safety logbook)
ISO 18436-2 Vibration analyst, cat. I or comparable	

3.2 Analysis personnel

Person 1
Vocational training in a technical field
ISO 18436-2 Vibration analyst, cat. II or comparable

4 The measuring equipment

The measuring equipment to be used must be fully inspected and calibrated.

4.1 Sensors

Requirements on vibration sensors

Acceleration sensors, piezo-resistive or capacitive
Linearity of the measuring chain up to (± 3 dB) 0.33 to 450 Hz
Sensitivity ≥ 100 mV/g ($\geq \sim 10$ mV/m/s ²)

Requirements on the speed sensor

Speed sensor based on the laser-optical or inductive principle (at least 12 pulses - at regular intervals - must be measured per turn)
Measuring range of the measuring chain starting at 0.1 Hz

4.2 Measuring equipment (portable system)

Measuring equipment with at least 2 channels
Sampling rate > 4000 Hz

5 Measurement

5.1 Operating conditions during measurement

Due to the experience gained with vibration measurements on the rotor bearings of ENERCON wind energy converters, the following operating conditions are required to perform a vibration measurement:

Stable production operations
No ice build-up
Permissible speed fluctuations, if possible, during measurement $\leq \pm 10$ %
No yaw movements during the measurement

When used, the acceleration sensors with piezo elements should be checked for their temperature limitations since these sensors often only function within a limited working temperature range.

5.2 Wind energy converter and bearing-specific data

The following specific wind energy converter and rotor bearing data may only be taken from official documents supplied by ENERCON.

Kinematic rolling-element bearing data (damage frequencies)

Ball pass frequency, inner race (BPFI)

Ball pass frequency, outer race (BPFO)

Spin frequency of rolling elements (BSF)

Cage rotation frequency (fundamental train frequency, FTF)

Distance of measuring points from flange between axle pin and main carrier

5.3 Installation and coupling of measurement system

1. Switch the WEC to maintenance mode.
2. Lock the rotor.
3. Sand off the paint coating of the axle pin at the measuring points shown in Figure 1 in the 12 o'clock position over an area of max. 4 x 4 cm approximately. Clean the work area thoroughly after sanding.
4. Fit the sensors in the positions marked in Figure 1 from which the paint coating has been removed.
 - When gluing, use hard glues based on cyanoacrylate.
 - If using a magnetic base, this must fit flush. Apply a thin layer of beeswax (or comparable) to improve coupling.

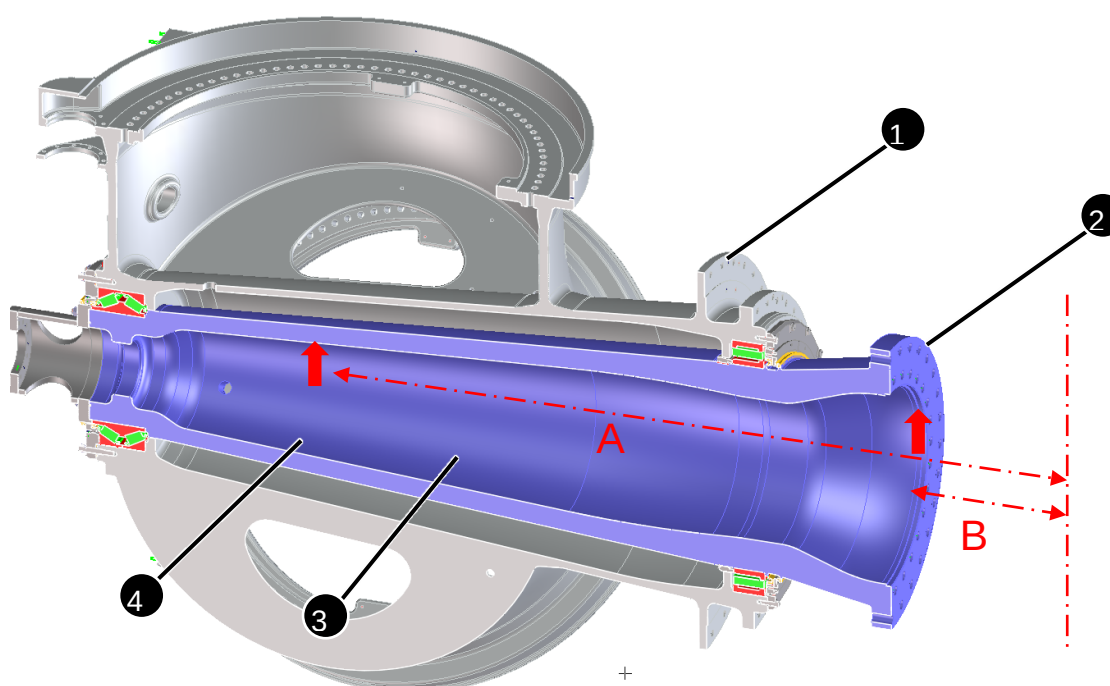


Fig. 1 Overview of measuring points on both rotor bearings, example of system type E70-E4

1	Hub	2	Rear rotor bearing
3	Axle pin	4	Front rotor bearing
A	Distance of front bearing from flange between axle pin and main carrier	B	Distance of rear bearing from flange between axle pin and main carrier

NOTICE



For measurements with magnetic base: Damage to sensors from vibration

Vibrations can cause damage to the sensors.

- ⇒ To protect the sensitive sensor, avoid unnecessary vibrations of the sensor head
- ⇒ Slowly rolling the sensor head along its edge onto the measuring surface is recommended.
- ⇒ The magnetic base of the sensor must lie flush on the measuring point and must not be allowed to wobble.



WARNING



Collision hazard when accessing and exiting the axle pin

Head injuries from colliding with the wall of the axle pin

- ⇒ Use PPE according to its intended use
- ⇒ A hard hat must be worn.
- ⇒ Check the vicinity for potential collision hazards.
- ⇒ Ensure adequate lighting in the axle pin.

5. Avoid allowing the sensor cables to vibrate. One way to prevent this is to attach the sensor cables to the wall of the axle pin using magnetic hooks.
6. Route the sensor cables in such a way that they are not in the immediate vicinity of live cables.



Live cables can affect the accuracy of the measurement if they are too close to the sensor cables routed in the axle pin.

7. Document the measurement set-up with photographs.
8. The speed must also be measured during the vibration measurement.
9. Once the installation of the measurement setup has been completed, release the rotor lock and allow the rotor to start rotating. Make sure that the speed sensor is not installed too close to the measuring point.
10. Start the vibration measurement as soon as stable production operation has been achieved.



During vibration measurement, the contents of the applicable operating instructions for the respective measurement system must be observed and followed.

11. During measurement, pitch control should be operated by a person authorised for switching or operation so as to minimise speed fluctuation during measurement at least to the extent required in chapter 5.1, but ideally to keep it as low as possible.
12. Observe the minimum measuring time for the vibration measurement!

RESTRICTED

5.4 Removal of the measurement system

13. Switch the WEC to maintenance mode.
14. Lock the rotor.
15. Remove the measuring equipment completely
16. Apply corrosion protection to the sanded metallic measuring points.

WARNING



Handling substances that are harmful to human health or allergenic (sensitising)

Poisoning, injury, allergies.

- ⇒ Wear suitable protective gloves according to the glove plan, protective clothing and safety glasses.
- ⇒ Avoid breathing in substances. Work only in well-ventilated rooms or use respiratory protection.
- ⇒ Follow all safety, storage and processing instructions provided by the respective manufacturer.
- ⇒ Observe the material safety data sheets for the individual substances.
- ⇒ Observe the latest applicable instructions for safe use of hazardous substances.

17. Mark the measurement positions unambiguously in colour (as reference points for future measurements)
18. Thoroughly clean the work area.
19. Release the rotor lock.
20. Start up the WEC.

6 Evaluation

6.1 Analysis of recorded vibration signals

The rotor speed of a wind energy converter is frequently subject to large fluctuations. As the damage frequencies of rolling-element bearings are speed-dependent, excessive speed fluctuations cause a distorted frequency image in frequency-selective analyses.

When performing vibration measurements on the rotor bearings of ENERCON wind energy converters, therefore, an order analysis must be carried out.

If the frequencies are analysed all the same, care must be taken to ensure that the speed fluctuations during measurement do not 'smear' the amplitudes in the spectrum.

7 Data required for the diagnostic report

To ensure that the employees of ENERCON Service are able to read and interpret the vibration diagnosis reports submitted and to initiate the correct actions, the following information must be provided in the vibration diagnosis reports. It is not essential to keep to the sequence shown:

- WEC data
 - Serial number of the WEC
 - Site
 - Nominal power
- Measurement details
 - Wind energy converter operator/owner
 - Customer with point of contact
 - Measuring person
 - Date and time
 - Wind speed during measurement
 - Speed during measurement
 - The generator output during measurement
 - The temperature of the RRB and FRB temp. sensor from the WEC control system
- Measuring equipment details
 - The measuring equipment
 - The sensors
 - Speed recording
- The measuring point arrangement
 - Photo documentation
- The kinematic data
- The conclusion
- The diagnostic results
- Analysis of the vibration measurement (graphical representations)

8 Checklist for the minimum requirements

WEC serial number		Site	
Measurement date		Evaluation date	

Subject area	Target	Actual				Comment
Qualifications of measuring personnel	ISO 18436-2 Vibration analyst, cat. I or comparable	Yes		No		
Qualifications of analysing personnel	ISO 18436-2 Vibration analyst, cat. II or comparable	Yes		No		
Measuring equipment - Sensors	Acceleration sensors, piezo-resistive or capacitive	Yes		No		
		Yes		No		
	Linearity of measurement chain up to 3 dB: 0.33 Hz to 450 Hz	Yes		No		
	Sensitivity ≥ 100 mV/g ($\geq \sim 10$ mV/m/s ²)	Yes		No		
Measuring equipment - Speed sensor	Speed sensor, laser-optical or inductive	Yes		No		
	Measuring range of the measuring chain starting at 0.1 Hz	Yes		No		
		Yes		No		
Measuring equipment - Measuring equipment	Measuring equipment with at least 2 channels	Yes		No		
	High sample rate > 4000 Hz	Yes		No		
		Yes		No		
		Yes		No		
Measurement conditions	Stable production operations	Yes		No		
	No ice build-up	Yes		No		

Subject area	Target	Actual				Comment
	Permissible speed fluctuations during measurement $\leq \pm 10\%$	Yes		No		
	No yaw movement during measurement	Yes		No		
	Measurement duration: ≥ 60 s. (at least 10 full turns of the rotor)	Yes		No		
	WEC is thermally in a steady state	Yes		No		
Wind energy converter and bearing-specific data	Ball pass frequency, inner race (BPFI)	VH			HV HRC	
	Ball pass frequency, outer race (BPFO)	VH			HV HRC	
	Spin frequency of rolling elements (BSF)	VH			HV HRC	
	Cage rotation frequency (fundamental train frequency, FTF)	VH			HV HRC	

Photos	
Left-click in one of the fields shown below to insert a photo.	
Fig. 1	Fig. 2
Fig. 3	Fig. 4

Fig. 5	Fig. 6
Fig. 7	Fig. 8
Fig. 9	Fig. 10